

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of B. Sc. Biological Sciences Examination March 2018

BS 302: Concepts of Organic Chemistry

Date: 31/03/2018 Day: Saturday Time: 10.00 a.m. to 10.20 a.m. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

- Q-I Choose the correct answer for the following questions. 10**
- 1 Hyperconjugation involves overlap of _____ orbitals.
(a) $\sigma - \sigma$ (b) $\sigma - p$ (c) $p - p$ (d) $\pi - \pi$
- 2 How many tertiary carbon atoms present in 2,4,4-trimethylhexane?
(a) One (b) Two (c) Three (d) Zero
- 3 Which of the following statement is NOT true for Fischer projection.
(a) we can rotate Fischer projection in plane 180°
(b) we cannot rotate Fischer projection in plane 270°
(c) we can rotate Fischer projection in plane 90°
(d) we cannot rotate Fischer projection out of plane 180°

- 4 Mesomeric effect operates only in the molecule having
- (a) double bond only (b) conjugated systems
(c) canonical structures (d) single bonds only
- 5 Elimination reaction involves.....
- (a) formation of one sigma bond and one pi bond.
(b) formation of two pi bonds.
(c) breaking of two pi bonds.
(d) breaking of two sigma bond and formation of new pi bond.
- 6 The shape of the $+CH_3$ is
- (a) Trigonal planar (b) Tetrahedral (c) Linear (d) square planar
- 7 What is the name of the alkane shown?
- 
- a) 3,3-dimethylhexane b) 2,2-dimethylhexane
c) 2-ethyl-2-methylpentane d) 2-methyl-2-ethylpentane
- 8 Nucleophilic aromatic substitution of aryl halides via addition-elimination path only proceeds when
- a) Electron withdrawing group present at meta to halide
b) Electron withdrawing group present at ortho and para to halide
c) Electron releasing group present at meta to halide
d) Electron releasing group present at meta to halide
- 9 Full form of IUPAC is
- (a) Important Union of Pure and Applied Chemistry
(b) International Union of Positive and Applied Chemistry
(c) International Union of Pure and Advanced Chemistry
(d) International Union of Pure and Applied Chemistry
- 10 Which of the following compound possess quaternary carbon?
- (a) 2-methylpentane (b) neopentane
(c) 2,3-dimethylhexane (d) isopentane

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of B. Sc. Biological Sciences Examination March 2018

BS 302: Concepts of Organic Chemistry

Date: 31/03/2018 Day: Saturday Time: 10.20 a.m. to 12.00 noon Maximum Marks: 25

Important Instructions:

- Section I and II must be attempted in SINGLE ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
-

SECTION-I

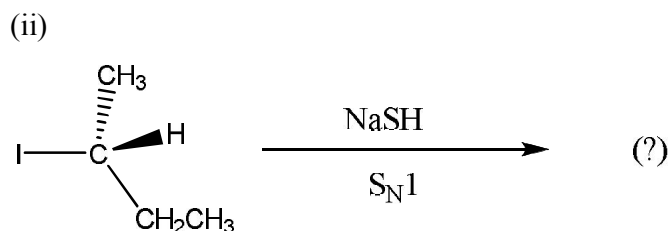
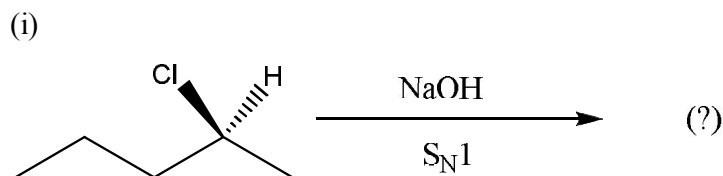
Q-II Answer the following questions as directed (Any Five). 10

- A Discuss the importance of stereochemistry.
- B Explain Inductive effect by giving suitable example.
- C Give the mechanism of halogenations of methane.
- D Discuss the Grignard reagent and its importance in organic chemistry.
- E Why atoms form bonds? Discuss your answer with suitable example.
- F How are organic compounds classified?

SECTION-II

Q-III Answer the following questions as directed (Any Five). 15

- A Complete and rewrite the following reaction and also predict the stereochemistry at chiral carbon.



- B** Write the elimination – Addition mechanism for nucleophilic aromatic substitution reaction.
- C** Why phenol undergoes nitration reaction very quickly than benzene? Explain your answer with detail mechanism.
- D** What makes carbon so special? Explain your answer in detail.
- E** Define the following terms with proper example.
(i) Enantiomer (ii) Homomer (iii) Meso compound
- F** Discuss the mechanism for alkylation of benzene with proper example.